Digital Electronic Circuits (EC2L004)

Lecture 0: Introduction & Logistics

Logistics

- Instructor

  Dr. Srinivas Boppu,
  School of Electrical Sciences, Room No: 106
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- Teaching Assistants

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  Ms. Mursina Khatun a23ec09005@iitbbs.ac.in
  Ms. Tiyasa Panja 22sp06004@iitbbs.ac.in
  Mr. Sachin Chauhan 23sp06007@iitbbs.ac.in
  Mr. Saikat Panja s24ec09003@iitbbs.ac.in
  Ms. Abhipsa Dutta 23sp06008@iitbbs.ac.in

- Lectures

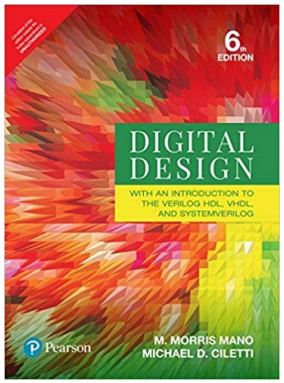
  Monday: 16:30 – 17:30, LBC 203
  Tuesday: 08:00 – 09:55, LBC 203

- Tutorial

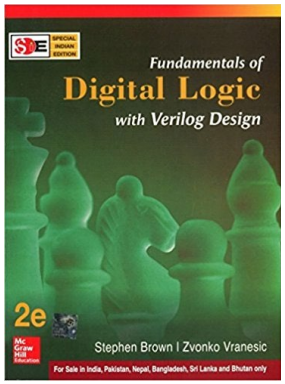
  Monday: 17:30 - 18.25 (LBC-203, ECE)
  Friday : 16:30 - 17:25 (LBC-203, CSE-1st Half)
  Friday : 17:30 - 18:25 (LBC-203, CSE-2nd Half)

Logistics: Cont'd…

- Text books
  - M.Morris Mano and Michael Ciletti, "Digital Design with an introduction to the Verilog HDL", Pearson publications.

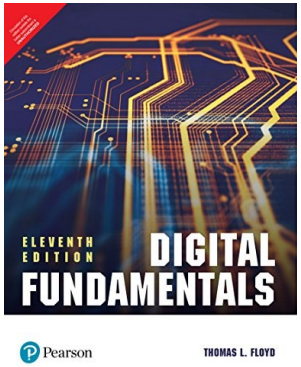


  - Stephen Brown and Zvonko Vranesic, "Fundamentals of Digital Logic with Verilog Design", McGraw Hill Education. Indian Edition.

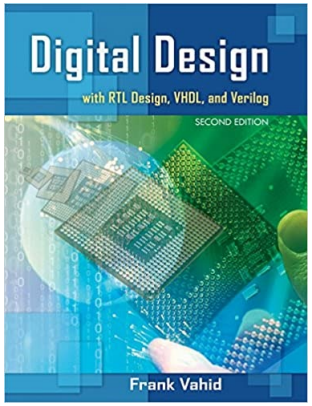


Logistics: Cont'd…

- Reference books
  - Thomas L. Floyd, "Digital Fundamentals", Pearson publications.

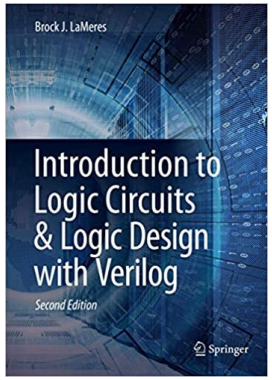


  - Frank Vahid, "Digital Design" Wiley Publications

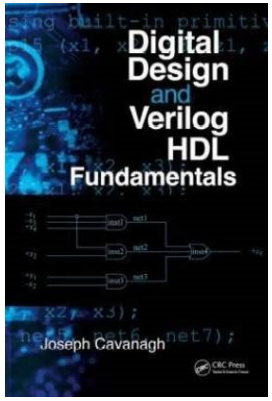


Logistics: Cont'd…

- Reference books
  - Brock J. LaMeres "Introduction to Logic Circuits & Logic Design with Verilog", Springer.

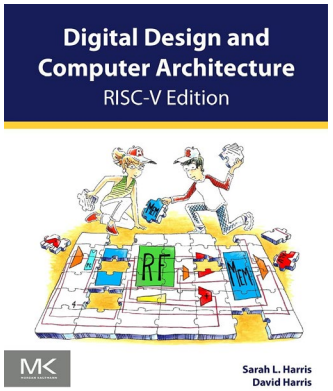


  - Joseph Cavanagh, "Digital Design and Verilog HDL Fundamentals", CRC press.

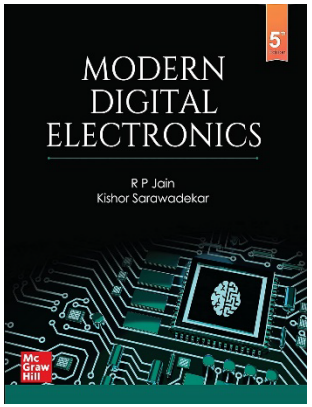


Logistics: Cont'd…

- Reference books
  - David Harris, Sarah L. Harris "Digital Design and Computer Architecture, RISC-V Edition.



  - R. P. Jain, Kishor Sarawadekar "Modern Digital Electronics" 5th Edition, McGrawHill.



Logistics: Cont'd…

- Grading:
  - Mid Semester: 30%
  - End Semester: 50%
  - Surprise Quizzes/ Assignments + Class Participation: 20%
  - Zero tolerance for copying solutions, cheating, and proxy for attendance.

Motivation: Digital Electronics

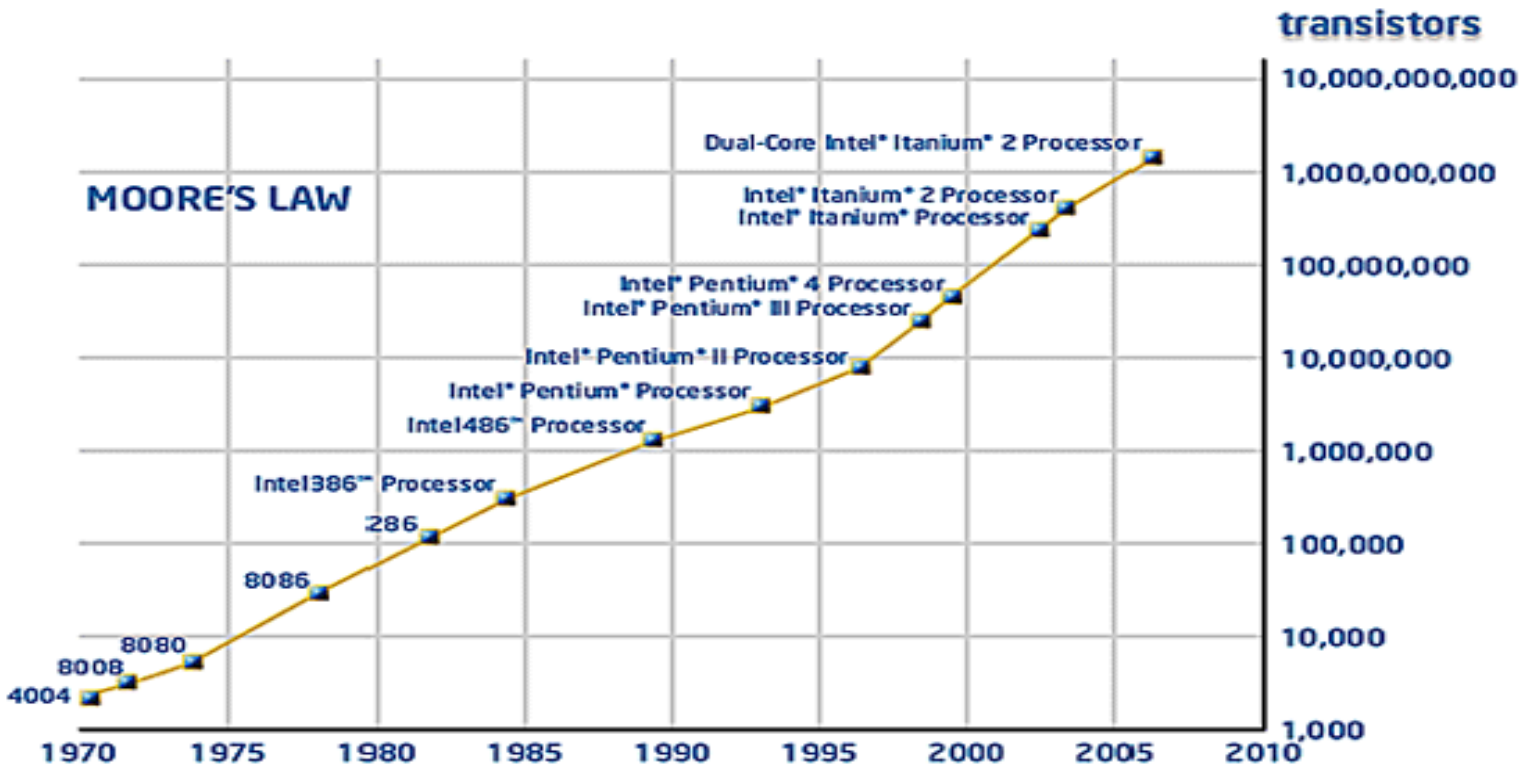
Discussion

- Digital Systems
  - Omnipresent in all the domains: communication, healthcare, industrial, entertainment, smart* etc.
  - May have a digital computer embedded in it
  - Interconnection of digital modules
- What is a digital computer?
  - A machine that follows a sequence of instructions, also called a program, that operates on a given data.

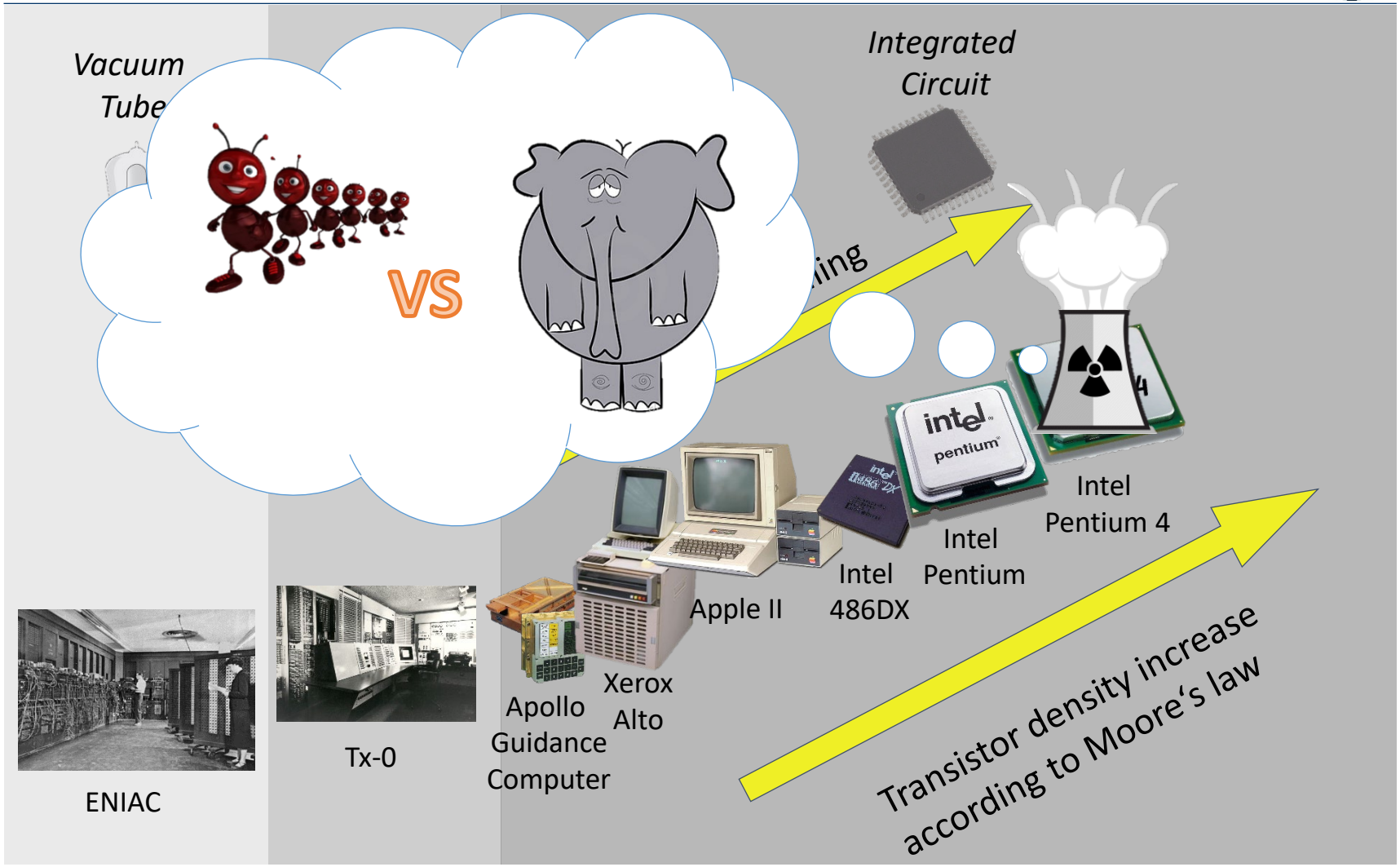
- What do *you* think this course is about?
  - Why is it important?
  - Fundamental course for Computer Architecture, VLSI Design (ASIC, FPGA), Electronic Design Automation etc.

Moore's Law

- Moore's law states that number of transistors in an integrated circuit doubles every 18 to 24 months.



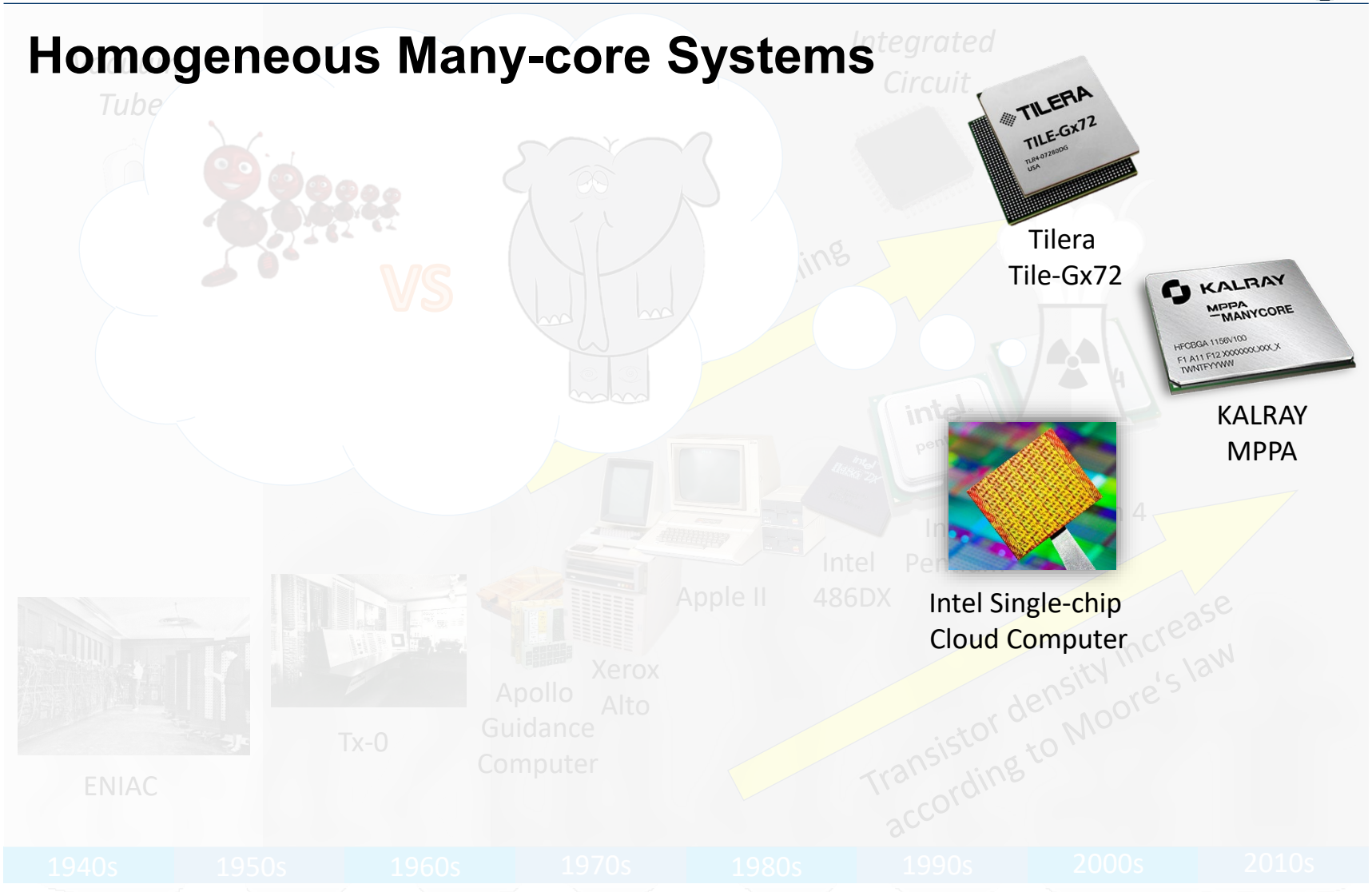
Source: intel



*Images from different sources on internet

Evolution of Integrated Circuits

Homogeneous Many-core Systems



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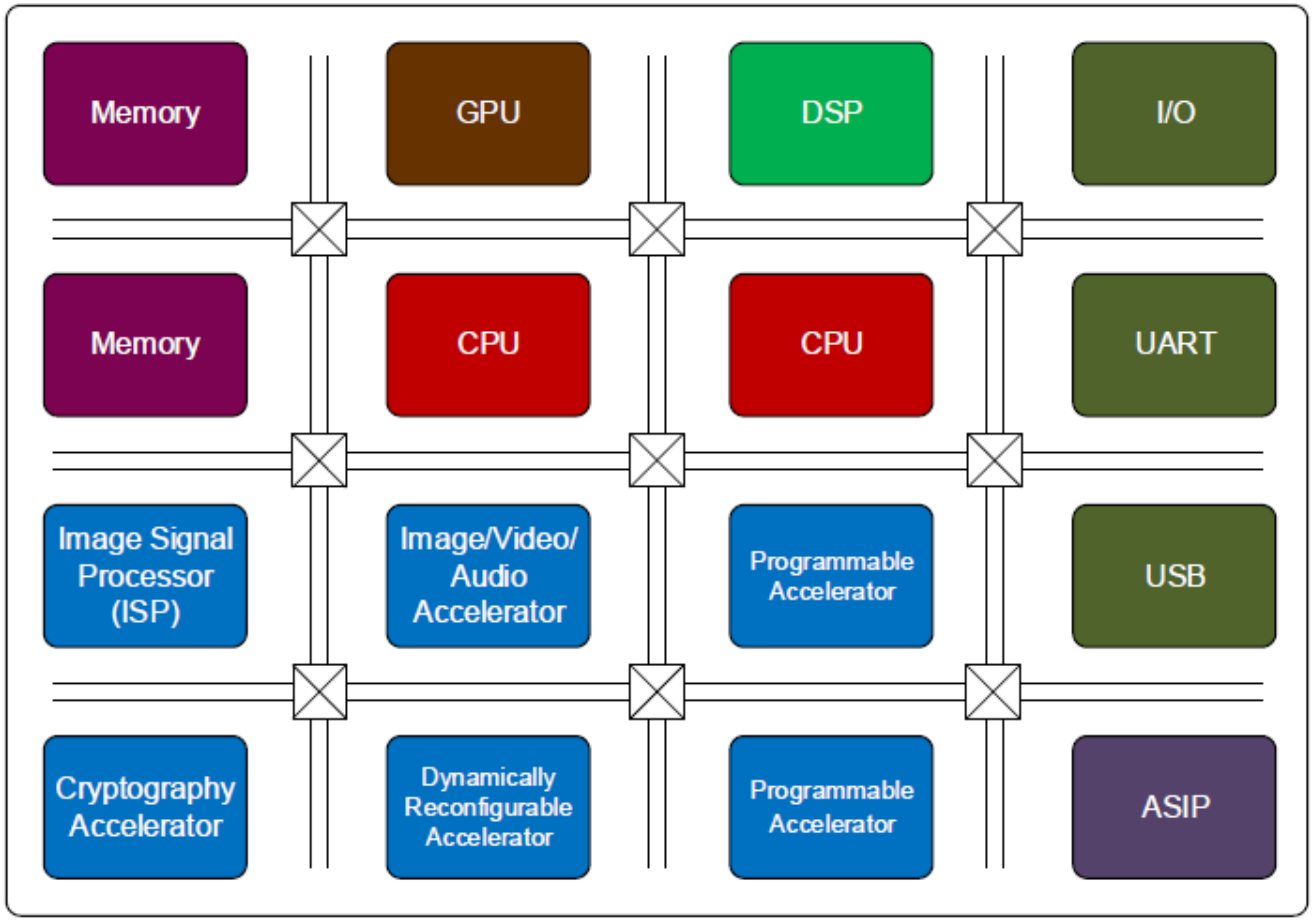
Heterogeneous Many-core Systems



In the future, only the customization and heterogeneity among the cores can improve energy efficiency the new fundamental limit for performance.

Course Outline

- Binary numbers: Number-base conversions
- Boolean algebra and logic gates
- Logic minimization
- Combinational Logic circuits
- Sequential logic circuits: storage elements, analysis of circuits, synthesis of circuits, registers, counters, and sequence detectors.
- Memory and programmable logic: RAM, ROM, PAL, and PLA.
- Asynchronous sequential logic : Analysis and design procedure
- Logic families: TTL and CMOS